TEXAS A&M UNIVERSITY Engineering

Patent Miner

Christian Casteel, Kosi Ezewudo, CJ VInes

Faculty Mentor: Dr. Lusher

TEXAS A&M UNIVERSITY Department of Electrical & Computer Engineering

Problem Definition

- Patent prior-art search relies heavily on keyword matching
- Similar inventions often use different terminology
- Missed results increase cost, time, and legal risk

Methodology

Patent Miner is a Retrieval Augmented Generation pipeline connected to a webpage consisting of the following subsystems:

Embedding Pipeline

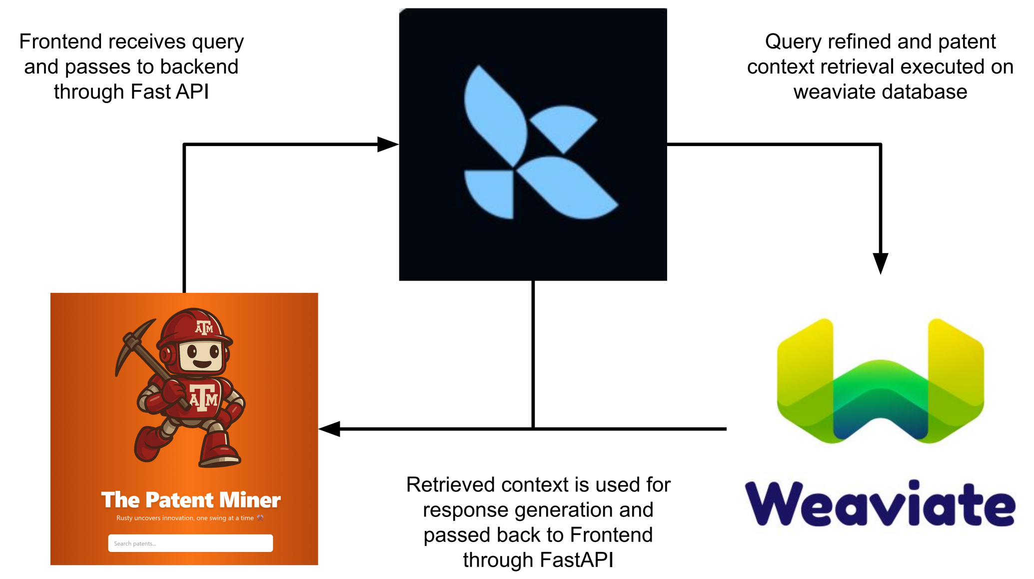
- Weaviate stores ColBERT claim embeddings and MUVERA fixed-length encodings
- Metadata indices for filtering and retrieval

Orchestrator

- Built with LangChain
- Handles query cleaning, routing, retrieval, and response generation using LLMs

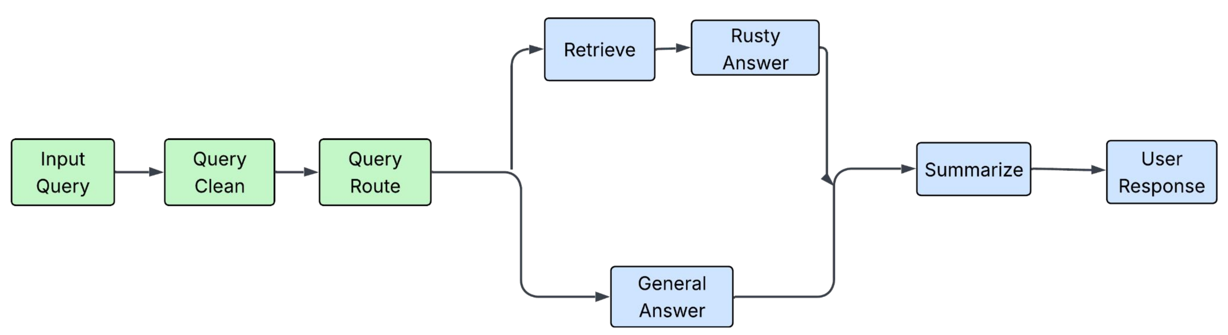
Frontend

- React + FastAPI web interface
- Accepts user queries and returns search/RAG responses

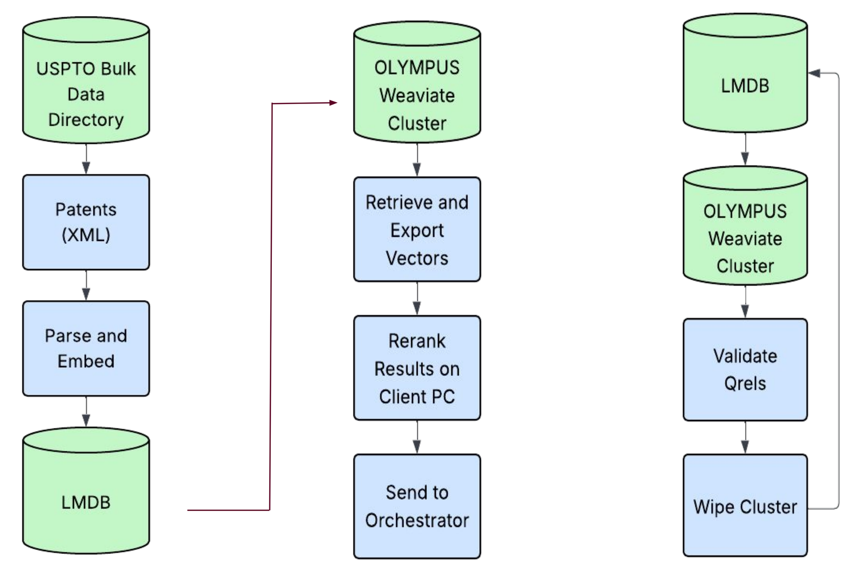


Engineering Analysis

Orchestrator Workflow



Embedding Pipeline Workflow



Retrieval Metrics

Top Results (k)	Patent Count	Recall@k	nDCG@k	MRR@k
10	~50,000	0.9	0.645	0.561
10	~100,000	0.8	0.589	0.520
10	~1,000,000	0.825	0.691	0.650

Sample Claim

1. A golf club head, comprising:

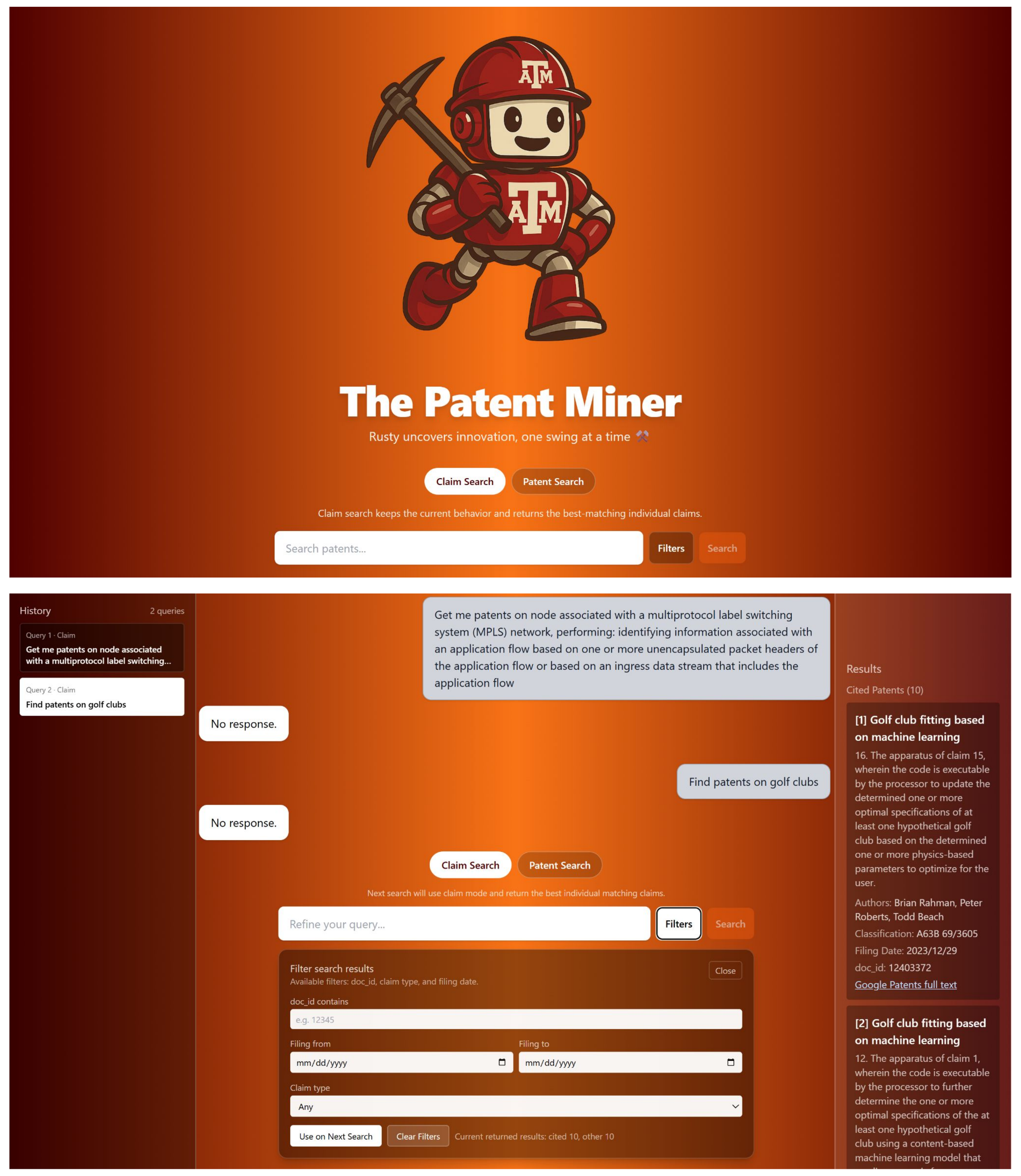
a body including a topline, a sole, and a front face, wherein the front face defines a rear surface that extends along a plane and the body defines a center of gravity plane, wherein the center of gravity plane extends parallel to a ground plane at a center of gravity of the body when the golf club head is at address; and

Sample Query

Additively manufactured golf club head with internal lattice or weight-tuning geometry to control center of gravity and mechanical response

Outcomes

- Patent Miner is a strong framework and working proof of concept for patent RAG systems.
- Successfully ingests patent data from the United States Patent and Trademark Office (USPTO)
- Conducts semantic and keyword based retrieval
- Outputs results through a web interface.before it can be considered scalable.



Future Work

Embedding Pipeline

- Improve scalability of the retrieval stack by migrating from Weaviate to FAISS + LMDB design.
- Fine tune ColBERT for patents so it can better capture claim and domain specific language.

Orchestrator

- Claim reconstruction can be implemented to further streamline the patent litigation process.
- The orchestrator could be trained to use the claims surfaced from a search to reconstruct target claims in the litigation process.

Impact

Patent Miner is a proof of concept tool that improves the workflow of patent litigation specialists without replacing them. It also helps users search for patents far more effectively than currently implemented keyword searches.

- **Litigation Tool:** Claim-level results so litigation specialists can find evidence more effectively.
- **Improved Search:** Semantic retrieval surfaces contextually relevant results as opposed to a keyword search.
- **Foundation for Future Systems:** Proves a working prototype for future patent RAG systems.

References

1. Lewis et al., Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks
2. Santhanam et al., ColBERTv2: Effective and Efficient Retrieval via Lightweight Late Interaction
3. Dhulipala et al., MUVERA: Multi-Vector Retrieval via Fixed Dimensional Encodings
4. Järvelin & Kekäläinen, Cumulated Gain-Based Evaluation of IR Techniques
5. LangChain RAG documentation
6. Weaviate multi-vector embeddings documentation

Acknowledgements

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